

Problem set 11 - Due: Wednesday, April 29

1. Use the second translation theorem

$$\mathcal{L}^{-1}\{e^{-as}F(s)\} = u(t-a)f(t-a)$$

to find the inverse of

$$F(s) = \frac{e^{-3s}}{s^2}.$$

We have

$$F(s) = e^{-3s} \frac{1}{s^2}.$$

Thus,

$$f(t) = \mathcal{L}^{-1}\left\{e^{-3s} \frac{1}{s^2}\right\}.$$

Let $G(s) = \frac{1}{s^2}$, then

$$g(t) = \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t.$$

So, $g(t-3) = t-3$, and

$$\mathcal{L}^{-1}\left\{e^{-3s} \frac{1}{s^2}\right\} = u(t-3)(t-3).$$

2. Use the second translation theorem

$$\mathcal{L}^{-1}\{e^{-as}F(s)\} = u(t-a)f(t-a)$$

to find the inverse of

$$F(s) = e^{-s} \frac{1}{s+2}.$$

Let $G(s) = \frac{1}{s+2}$, then

$$g(t) = \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = e^{-2t}.$$

Since $a = 1$, $g(t-1) = e^{-2(t-1)}$. Therefore,

$$\mathcal{L}^{-1}\left\{e^{-s} \frac{1}{s+2}\right\} = u(t-1)e^{-2(t-1)}.$$

3. Use the second translation theorem

$$\mathcal{L}^{-1} \{e^{-as} F(s)\} = u(t-a)f(t-a)$$

to find the inverse of

$$F(s) = e^{-s} \frac{s}{s^2 + \pi^2}.$$

Let $G(s) = \frac{s}{s^2 + \pi^2}$, then

$$g(t) = \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + \pi^2} \right\} = \cos(\pi t).$$

Then, since $a = 1$,

$$f(t) = u(t-1) \cos(\pi(t-1)).$$

3. Use Laplace transform method to solve the ODE

$$x'' + 4x = \begin{cases} \cos(2t) & 0 \leq t < 2\pi \\ 0 & t \geq 2\pi \end{cases}, x(0) = x'(0) = 0.$$

We have

$$\begin{cases} \cos(2t) & 0 \leq t < 2\pi \\ 0 & t \geq 2\pi \end{cases} = \begin{cases} h & 0 \leq t < 2\pi \\ h & t \geq 2\pi \end{cases} + \begin{cases} 0 & 0 \leq t < 2\pi \\ f(t-2\pi) & t \geq 2\pi \end{cases}.$$

Thus,

$$\begin{aligned} \cos(2t) &= h, \\ 0 = \cos(2t) + f(t-2\pi) &\implies f(t-2\pi) = -\cos(2t). \end{aligned}$$

So,

$$\begin{cases} \cos(2t) & 0 \leq t < 2\pi \\ 0 & t \geq 2\pi \end{cases} = \cos(2t) - \cos(2t)u(t-2\pi).$$

Then,

$$\begin{aligned} \mathcal{L}\{\cos(2t)\} - \mathcal{L}\{\cos(2t)u(t-2\pi)\} &= \frac{s}{s^2 + 4} - e^{-2\pi s} \frac{s}{s^2 + 4} \\ &= (1 - e^{-2\pi s}) \frac{s}{s^2 + 4}. \end{aligned}$$

So, we have

$$\begin{aligned} \mathcal{L}\{x'' + 4x\} &= (1 - e^{-2\pi s}) \frac{s}{s^2 + 4} \\ \implies s^2 X(s) - sx(0) - x'(0) + 4X(s) &= (1 - e^{-2\pi s}) \frac{s}{s^2 + 4} \\ \implies (s^2 + 4)X(s) &= (1 - e^{-2\pi s}) \frac{s}{s^2 + 4} \\ \implies X(s) &= (1 - e^{-2\pi s}) \frac{s}{(s^2 + 4)^2}. \end{aligned}$$

Thus,

$$X(s) = \frac{s}{(s^2 + 4)^2} - e^{-2\pi s} \frac{s}{(s^2 + 4)^2},$$

so

$$x(t) = \mathcal{L}^{-1} \left\{ \frac{s}{(s^2 + 4)^2} \right\} - \mathcal{L}^{-1} \left\{ e^{-2\pi s} \frac{s}{(s^2 + 4)^2} \right\}.$$

With

$$\mathcal{L}^{-1} \left\{ \frac{s}{(s^2 + 4)^2} \right\} = \frac{1}{4} t \sin(2t),$$

we let $G(s) = \frac{s}{(s^2 + 4)^2}$, so

$$g(t) = \frac{1}{4} t \sin(2t).$$

With $a = 2\pi$, $g(t - 2\pi) = \frac{1}{4}(t - 2\pi) \sin(2(t - 2\pi))$, so

$$x(t) = \frac{1}{4} t \sin(2t) - u(t - 2\pi) \frac{1}{4} (t - 2\pi) \sin(2(t - 2\pi)).$$